

CV – Dr. Magali Frauendorf

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Sweden

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29-04-1988, Eckernförde, Germany
German and French nationality



EDUCATION

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|-----------|---|
| 2016-2021 | PhD, Department of Animal Ecology, Netherlands Institute of Ecology (NL) |
| 2014-2016 | Master of Science in Forest and Nature Conservation, Wageningen University, Wageningen (NL) (cum laude) |
| 2008-2012 | Bachelor of Applied Science in Wildlife Management, University of Applied Sciences, Leeuwarden (NL) |



RESEARCH EXPERIENCE

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|---|-----------------------|
| Department of Wildlife, Fish and Environmental Studies, Swedish University of Agricultural Sciences, SE
Post-doctoral researcher
<i>Project & tasks:</i> Applying technologies around camera trapping and AI as well as citizen science to study ecological questions related to wildlife conservation and management, student supervision. | Feb 2023 -
now |
| Altenburg & Wymenga ecological research
Ecological researcher
<i>Project & tasks:</i> Fieldwork, student supervision, project leading, data analysis, report writing in project concerning meadow birds, shorebirds, small mustelids and geese management. | Jan 2022-
Jan 2023 |
| Department of Animal Ecology, Netherlands Institute of Ecology, NL
PhD student
<i>Project:</i> "Causes of spatiotemporal variation in reproductive performance of Eurasian oystercatchers in a human-dominated landscape"; https://zenodo.org/record/6837142
<i>Tasks:</i> Fieldwork (Monitoring breeding success of oystercatchers; catching, ringing, gps tagging of oystercatchers in the Wadden Sea and Delta area; measuring body condition (physiology and morphology); citizen science; data analysis (Structural Equation Modeling, Bayesian Statistics, Mark-Recapture)
<i>Supervision:</i> Martijn van de Pol, Bruno J. Ens, Eelke Jongejans, Hans de Kroon | Dec 2016-
Dec 2021 |
| Institute for Terrestrial and Aquatic Wildlife Research, University of Veterinary Medicine Hannover, Germany
Research assistant
<i>Project:</i> "Space use of red deer in Lower Saxony" and "Investigations on the influence of hunting on geese and crop damages"
<i>Tasks:</i> writing project proposal for financial funding, doing data analysis in GIS and R as well as field work (catching and collaring geese).
<i>Supervisor:</i> Oliver Keuling | May 2016-
Dec 2016 |

**Wageningen Environmental Research, Wageningen, NL**

Nov 2015-
March 2016

Intern (master)

Project: "Investigating the effectiveness of different non-invasive methods (trap cam, video surveillance and track pads) to encounter the usage of crossing structures by ungulates in the National Park Hoge Veluwe."

Tasks: field work (trap cam and track pad monitoring), statistical analysis, report writing

Supervisor: Sip van Wieren, Edgar van der Grift

Wageningen University, NL & Institute for Terrestrial and Aquatic Wildlife Research, Germany

Feb 2015-
July 2015

Thesis (master)

Project: "The influence of environmental factors, physiological factors and hunting on the reproductive success of wild boar in Germany"

Tasks: Statistical analysis of long-term dataset of wild boar reproduction and writing up results in three manuscripts resulting in one published paper.

Supervision: Sip van Wieren, Oliver Keuling

Institute for Terrestrial and Aquatic Wildlife Research, Germany

April 2013-
Dec 2013

Research assistant

Project: "Wild boar management in Lower Saxony"

Tasks: telemetry and catching of wild boar & habitat analysis

Supervision: Oliver Keuling

Norwegian Institute for Nature Research (NINA), Norway

June 2012-
Aug 2012

Thesis (bachelor)

Project: "Microhabitat characteristics of Eurasian Lynx (*Lynx lynx*) kill and rest sites in northern Norway"

Tasks: GPS-tracking of lynx, finding kill and rest sites, measuring microhabitat characteristics

Supervisors: Jenny Mattisson, John Linnell, John Odden

Harz National Park (Lynx project), Germany

Sep 2011-
Feb 2012

Intern (bachelor)

Project: "Pilot study on the microhabitat selection of Eurasian lynx (*Lynx lynx*) concerning hunting sites in the Harz National Park, Germany"

Tasks: catching lynx, following lynx with VHF and GPS tracking, finding kill and rest sites, following tracks in the snow, statistical analysis, report writing.

Supervisor: Ole Anders

Research and education centre „La Tirimbina“, Costa Rica

Feb 2010-
June 2010

Intern (bachelor)

Project: "Density estimation of Costa Rican wet tropical forest bird species"

Tasks: GIS-analysis, mist netting bats, bird monitoring.

Supervisor: Melquisedec Gamba-Rios

**Published (chronologically ordered):**

- 1) van de Pol, M., Bailey, L. D., **Frauendorf, M.**, Allen, A. M., van der Sluijs, M., Hijner, N., Brouwer, L., de Kroon, H., Jongejans, E., & Ens, B. J. (2024). Sea-level rise causes shorebird population collapse before habitats drown. *Nature Climate Change*, 14(8), 839–844. <https://doi.org/10.1038/s41558-024-02051-w>
- 2) Oosterveld, E. B., & **Frauendorf, M.** (2024). Is the settlement of a winter territory in the European Robin *Erithacus rubecula* triggered by the arrival of conspecific migrants? *Journal of Ornithology*, 1–6. <https://doi.org/10.1007/s10336-024-02150-7>
- 3) **Frauendorf, M.**, Allen, A. M., Jongejans, E., Ens, B. J., Teunissen, W., Kampichler, C., van Turnhout, C. A. M., Bailey, L. D., de Kroon, H., Cremer, J., Kleyheeg, E., Nienhuis, J., & van de Pol, M. (2022). Love thy neighbour?—Spatial variation in density dependence of nest survival in relation to predator community. *Diversity and Distributions*, 28(4), 624–635. <https://doi.org/10.1111/ddi.13457>
- 4) van Irsel, J., **Frauendorf, M.**, Ens, B. J., van de Pol, M., Troost, K., Oosterbeek, K., de Kroon, H., Jongejans, E., & Allen, A. M. (2022). State-dependent environmental sensitivity of reproductive success and survival in a shorebird. *Ibis*, 164(3), 692–710. <https://doi.org/10.1111/ibi.13038>
- 5) Allen, A. M., Jongejans, E., van de Pol, M., Ens, B. J., **Frauendorf, M.**, van der Sluijs, M., & de Kroon, H. (2022). The demographic causes of population change vary across four decades in a long-lived shorebird. *Ecology*, 103(4), e3615. <https://doi.org/10.1002/ecy.3615>
- 6) Kolk, H.-J. van der, Desmet, P., Oosterbeek, K., Allen, A. M., Baptist, M. J., Bom, R. A., Davidson, S. C., Jong, J. de, Kroon, H. de, Dijkstra, B., Dillerop, R., Dokter, A. M., **Frauendorf, M.**, Milotić, T., Rakhimberdiev, E., Shamoun-Baranes, J., Spanoghe, G., Pol, M. van de, Ryckegem, G. V., ... Ens, B. J. (2022). GPS tracking data of Eurasian oystercatchers (*Haematopus ostralegus*) from the Netherlands and Belgium. *ZooKeys*, 1123, 31–45. <https://doi.org/10.3897/zookeys.1123.90623>
- 7) **Frauendorf, M.**, Allen, A. M., Verhulst, S., Jongejans, E., Ens, B. J., van der Kolk, H.-J., de Kroon, H., Nienhuis, J., & van de Pol, M. (2021). Conceptualizing and quantifying body condition using structural equation modelling: A user guide. *Journal of Animal Ecology*, 90(11), 2478–2496. <https://doi.org/10.1111/1365-2656.13578>
- 8) van der Kolk, H.-J., Ens, B. J., **Frauendorf, M.**, Jongejans, E., Oosterbeek, K., Bouten, W., & van de Pol, M. (2021). Why time-limited individuals can make populations more vulnerable to disturbance. *Oikos*, 130(4), 637–651. <https://doi.org/10.1111/oik.08031>
- 9) van der Kolk, H.-J., Krijgsveld, K. L., Linssen, H., Diertens, R., Dolman, D., Jans, M., **Frauendorf, M.**, Ens, B. J., & van de Pol, M. (2020). Cumulative energetic costs of military aircraft, recreational and natural disturbance in roosting shorebirds. *Animal Conservation*, 23(4), 359–372. <https://doi.org/10.1111/acv.12546>
- 10) Allen, A. M., Ens, B. J., van de Pol, M., van der Jeugd, H., **Frauendorf, M.**, Oosterbeek, K., & Jongejans, E. (2019). Seasonal survival and migratory connectivity of the Eurasian Oystercatcher revealed by citizen science. *Auk*, 136(1), 1–17. <https://doi.org/10.1093/auk/uky001>
- 11) Allen, A. M., Ens, B. J., van de Pol, M., van der Jeugd, H., **Frauendorf, M.**, van der Kolk, H.-J., Oosterbeek, K., Nienhuis, J., & Jongejans, E. (2019). Colour-ring wear and loss effects in citizen science mark-resighting studies. *Avian Research*, 10(11), 1–15. <https://doi.org/10.1186/s40657-019-0151-z>
- 12) Linssen, H., van de Pol, M., Allen, A. M., Jans, M., Ens, B. J., Krijgsveld, K. L., **Frauendorf, M.**, & van der Kolk, H.-J. (2019). Disturbance increases high tide travel distance of a roosting shorebird but only marginally affects daily energy expenditure. *Avian Research*, 10(31), 1–11. <https://doi.org/10.1186/s40657-019-0171-8>

**Published (chronologically ordered):**

- 13) van der Kolk, H.-J., Ens, B. J., Oosterbeek, K., Bouten, W., Allen, A. M., **Frauendorf, M.**, Lameris, T. K., Oosterbeek, T., Deuzeman, S., de Vries, K., Jongejans, E., & van de Pol, M. (2019). Shorebird feeding specialists differ in how environmental conditions alter their foraging time. *Behavioral Ecology*, 31(2), 371–382. <https://doi.org/10.1093/beheco/arz189>
- 14) **Frauendorf, M.**, Gethöffer, F., Siebert, U., & Keuling, O. (2016). The influence of environmental and physiological factors on the litter size of wild boar (*Sus scrofa*) in an agriculture dominated area in Germany. *Science of the Total Environment*, 541, 877–882. <https://doi.org/10.1016/j.scitotenv.2015.09.128>

Submitted or under review/revision (alphabetically ordered):

- Catfolis, B., **Frauendorf, M.**, Verheyen, K., Baeten, L., Vanroy, T., Martel, A., Pasmus, F., Verbrugghe, E., Lens, L., & Strubbe, D. Bird health in forests: exploring relationships between forest complexity, body condition, and tick infestation. *Heliyon Environment*.
- **Frauendorf, M.**, Allen, A. M., van der Kolk, H.-J., Cubaynes, S., Ens, B. J., Verhulst, S., Jongejans, E., de Kroon, H., Oosterbeek, K., Troost, K., & Van de Pol, M. Body condition mediates carry-over effects of a deteriorating winter habitat on reproduction in a declining shorebird. *Ecological Applications*.
- **Roilo, S.**, Hofmeester, T. R., Frauendorf, M., Widén, A., & Cord, A. The untapped potential of camera traps for farmland biodiversity monitoring: current practice and outstanding agroecological questions. *Remote Sensing in Ecology and Conservation*.

In preparation:

- Osinga, T., **Frauendorf, M.**, de Knecht, H. J., Hofmeester, T. R. Setting Thresholds: Minimal detection probabilities for reliable multi-state occupancy models. *Ecosphere*.
- **Frauendorf, M.**, Ens, B. J., Allen, A. M., Jongejans, E., de Kroon, H., Teunissen, W., Kampichler, C., Kleyheeg, E., Nienhuis, J., Oosterbeek, K., van de Sluijs, M., de Jong, M., van de Pol, M. The synergistic effects of environmental drivers of declining reproduction of a threatened meadow bird across the Netherlands. *Journal of Applied Ecology*.
- **Frauendorf, M.**, et al. Trapper Citizen Science: a platform to promote citizen science engagement for ecological research and wildlife management through camera trapping.
- **Frauendorf, M.**, et al. Using computer vision for a quick estimation of ungulate reproduction from camera trap data.



CONFERENCE PARTICIPATION

- 2024 *Viltkonferensen. Stockholm, Sweden. Oral presentation.*
- 2024 *Nordic Oikos Conference. Lund, Sweden. Oral presentation. Co-hosted session.*
- 2024 *Camera Traps, AI, and Ecology. Online.*
- 2023 *Forum for CV in Ecology and Evolutionary Biology. Lund, Sweden. Oral presentation*
- 2021 *International Wader Study Group Conference (IWSG). Online.*
- 2020 *International Wader Study Group Conference (IWSG). Online. Oral presentation.*
- 2020 *III Elvonal Shorebird Conference. Debrecen, Hungary. Oral presentation.*
- 2020 *Netherlands Annual Ecology Meeting (NAEM). Lunteren, the Netherlands. Oral presentation.*
- 2019 *British Ecological Society Meeting (BES). Belfast, Ireland. Oral presentation.*
- 2019 *Netherlands Annual Ecology Meeting (NAEM). Lunteren, the Netherlands. Co-hosted session.*
- 2019 *European ornithological Union conference (EOU). Cluj-Napoca, Romania. Oral presentation.*
- 2018 *International Wader study group conference (IWSG). Workum, Netherlands. Oral presentation.*
- 2018 *International ornithological society conference (IOC). Vancouver, Canada. Oral presentation.*
- 2017 *Netherlands Annual Ecology Meeting (NAEM). Lunteren, the Netherlands. Attendance.*
- 2015 *Annual Meeting of the German Society for Mammalian Biology (DGS). Hannover, Germany. Oral presentation.*



REVIEW EXPERIENCE

- Reviewed in Animal Biotelemetry, Biological Conservation, Ecology and Evolution, European Journal of Wildlife Research, Global Change Biology, Journal of Animal Ecology, PeerJ, Royal Society Open Science, Scientific Reports, Wader study (ordered alphabetically)
- Editor: I am member of the editorial board of the scientific Dutch Ornithological Union journal Ardea.



SUPERVISION & TEACHING EXPERIENCE

- Supervision of thirteen Bachelor and Master students (2017-2024)
- Three guest lectures in course *Conservation Biology* for Bachelor and Master students at Wageningen University and Radboud University (2017, 2018)
- Lecture in the *Course Census Methods for Biodiversity (VFM)* on AI methods for biodiversity monitoring (2024)



SKILLS & COMPETENCES

Field work	Catching different species (wild boar, lynx, bats, birds) with different techniques (mist net, cannon net, trap box) • ringing birds • monitoring reproductive success • following animals with gps/telemetry/colour ring/(snow) tracks • trap cam monitoring • bat monitoring (bat detector) • citizen science
Software	Python • ArcGis • Qgis • R • JAGS • SPSS • Distance • Ranges • Maxent • Vortex • Matlab • Endnote • Zotero • Batsound
Languages	German: fluent • English: fluent • Dutch: fluent • French: fluent • Swedish: basic (level B1) • Spanish: basic
Other	Valid driving licence class BE (car-trailer, since 2006)



PARTICIPATED COURSES

- Supervising doctoral students planned for spring 2025
- CV4Ecology summer course
- Teaching in higher education
- Grading and assessment
- Structural Equation Modelling
- Integral Projection Models
- Applied Spatial Modelling with R
- Modelling Population Dynamics – Estimating Demographic Parameters for Wildlife Conservation
- Bayesian inference of animal demography
- The Floor is Yours - How to present/pitch your research online
- Make me fly - Stress management during your PhD



MAJOR COLLABORATORS

- Andrew Allen, Netherlands Institute of Ecology, Wageningen, NL
- Bruno Ens, Sovon Dutch centre for field ornithology, Nijmegen, NL
- Christian Kampichler, Sovon Dutch centre for field ornithology, Nijmegen, NL
- Eelke Jongejans, Radboud University, Nijmegen, NL
- Filip Ånöstam, Svenska Jägareförbundet, Öster Malma, SE
- Fredrik Widemo, Department of Wildlife, Fish and Environmental Studies (SLU), Umeå, SE
- Hans de Kroon, Radboud University, Nijmegen, NL
- Jakub Bubnicki, Mammal Research Institute, Polish Academy of Science, Open Science Conservation Fund, Białowieża, PL
- Joris Cromsigt, Department of Wildlife, Fish and Environmental Studies (SLU), Umeå, SE
- Liam Bailey, Leibniz Institute for Zoo and Wildlife Research, Berlin, GER
- Martijn van de Pol, Netherlands Institute of Ecology, Wageningen, NL/ James Cook University, Townsville, AUS
- Oliver Keuling, Institute for Terrestrial and Aquatic Wildlife Research, Hannover, GER
- Simon Verhulst, University of Groningen, Groningen, NL
- Sarah Cubaynes, Centre for Functional and Evolutionary Ecology, Montpellier, FR
- Simon Chamaillé-Jammes, Centre for Functional and Evolutionary Ecology, Montpellier, FR
- Tim Hofmeester, Department of Wildlife, Fish and Environmental Studies (SLU), Umeå, SE
- Vincent Miele, Alpine Ecology Laboratory, Chambéry, FR
- Wolf Teunissen, Sovon Dutch centre for field ornithology, Nijmegen, NL



NATIONAL SOCIETALLY RELEVANT OUTREACH ACTIVITY

- Scholeksters op daken; Natuurlijk Zoey, RTV Utrecht, 2018.
- Pilot broedsucces metingen van de scholekster in Noord-Holland; Jaarboek Boerenlandvogels Noord-Holland; 2018.
- Bedreiging scholekster komt van alle kanten; Artikel in Bionieuws; 2017.
- Interessant en dramatisch broedseizoen voor Nederlandse scholeksters; NatureToday; 2017.
- Scholeksters zoeken-Hoe komt het dat de scholekster zo fel achteruit gaat? Mens&Vogel. 55: 48-57; 2017.
- Interview op Omrop Fryslân; 2017.
- Speur dit voorjaar mee naar scholeksters met kleurringen; NatureToday; 2017.



“OTHER” EMPLOYMENT/EXPERIENCE

2017-2020	Volunteer in the breeding bird monitoring program (BMP); Sovon, Dutch Centre for Field Ornithology.
2018-present	Collaborator of the Élvonal Shorebird Science project (www.elvonashorebirds.com)
2018-2021	Assisting in research activities in the citizen science project ('Scholekster op het dak') about roof-breeding oystercatchers (www.scholeksterophetdak.nl)